

INTERNSHIP PROJECT REPORT TASK-4

Title

Create account on the below domain
(link: <https://cyberdefenders.org/blueteam-ctf-challenges>) solve the HawkEye lab. Note every step and tool you used to solve this lab and create a detailed report on it.

1. Introduction

As part of my cybersecurity internship program, I was assigned the task of creating an account on the CyberDefenders platform and solving the HawkEye Blue Team challenge. The objective of this project was to perform a practical network forensic investigation by analyzing a packet capture (PCAP) file and identifying the activities of a credential-stealing malware.

During this project, I downloaded the provided PCAP file and analyzed it using Wireshark on Kali Linux. I examined different types of network traffic, including DNS, HTTP, TCP, and SMTP communications, to understand the behaviour of the malware. By investigating the captured packets, I was able to identify malicious domains, IP addresses, email communications, stolen credentials and other important evidence related to the attack.

2. Background

Cyber-attacks are becoming more common, and attackers often use malware to steal sensitive information from users. This information can include usernames, passwords, banking details, and email credentials. To investigate such incidents, cybersecurity professionals analyse network traffic and system activity to understand what happened and how the attack was carried out.

Using Wireshark on Kali Linux, I examined different network communications and tracked the actions of the malware. During the investigation, I identified suspicious domains, IP addresses, downloaded files, email communications and stolen credentials. This project helped me gain practical experience in network traffic analysis, malware investigation and digital forensics while understanding how security analysts handle real-world cyber incidents.

3. Learning Objectives

The main aim of this project was to learn how network traffic analysis is used to investigate cybersecurity incidents. Through the HawkEye challenge, I wanted to gain practical experience in analysing captured network traffic and understanding how malware communicates within a network.

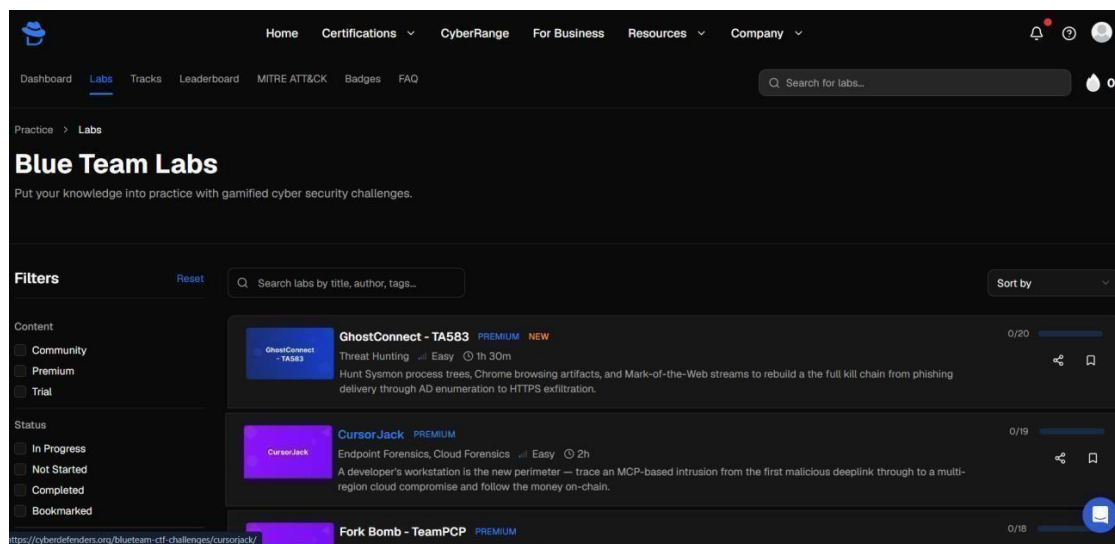
The main objectives of this project were:

- To understand the basics of network forensic investigation.
- To learn how to use Wireshark for analysing packet capture (PCAP) files.
- To identify suspicious network activities and malicious communications.
- To learn how to investigate DNS, HTTP, TCP, and SMTP network traffic.
- To identify malicious domains, IP addresses, and other indicators of compromise (IOCs).
- To understand how attackers steal sensitive information from infected systems.

4. Activities and Tasks Performed

Step1 -Creating a CyberDefenders Account

I began the project by visiting the CyberDefenders platform and creating a personal account. After completing the registration process, I logged in to the platform and explored the Blue Team Labs section. This gave me access to various cybersecurity challenges designed to improve practical investigation skills.



Step2 -Accessing the HawkEye Lab and Downloading the Files

After successfully logging into CyberDefenders, I searched for the HawkEye Lab in the Blue Team Labs section and opened the challenge page. I carefully reviewed the lab description, scenario, objectives, and the tools recommended for the investigation. The scenario explained that suspicious network activity was detected after an employee opened a malicious email attachment and my task was to investigate the incident.

The screenshot shows the CyberDefenders interface for the HawkEye Lab. The top navigation bar includes links for Dashboard, Labs (active), Tracks, Leaderboard, MITRE ATT&CK, Badges, and FAQ. A search bar is present with the text "Search for labs...". The main header shows the path: Practice > SOC Analyst Tier 2 > Level 6 > HawkEye. The lab title "HawkEye Lab" is prominently displayed. Below the title, a description states: "Reconstruct a HawkEye Keylogger data exfiltration incident by analyzing network traffic with Wireshark and CyberChef, Identifying IoCs and stolen credentials." The category is "Network Forensics". Tactics listed include Initial Access, Execution, Defense Evasion, Credential Access, Discovery, Collection, Command and Control, and Exfiltration. Tools listed include Wireshark, Brim, Apackets, MaxMind Geo IP, and VirusTotal. The lab is marked as "Retired" and has a rating of 4.5 stars. Action buttons for Bookmark, Join the Lab Squad, and Report an Issue are visible. On the left, there is a "Download Lab Files" button and a note: "Unzip the file with password cyberdefenders.org". Below this, it says "For a safe experience, consider opening this content in a secure, isolated environment." At the bottom left, it shows "0 / 24 Questions" and "0% Completed". On the right, the "Scenario" section is expanded, showing instructions: "Uncompress the lab (pass: cyberdefenders.org)". The scenario text reads: "An accountant at your organization received an email regarding an invoice with a download link. Suspicious network traffic was observed shortly after opening the email. As a SOC analyst, investigate the network trace and analyze exfiltration attempts."

Step 3 - Preparing the Investigation Environment

Before starting the analysis, I launched my Kali Linux virtual machine using Oracle VirtualBox. Kali Linux was chosen because it contains the tools required for network traffic analysis and digital forensic investigations.



Step 4 - Creating a Project Workspace

I created a folder named HawkEye in Kali Linux using terminal commands. I then opened the folder and checked its location to confirm it was created successfully. This folder was used to store all files related to the HawkEye lab.

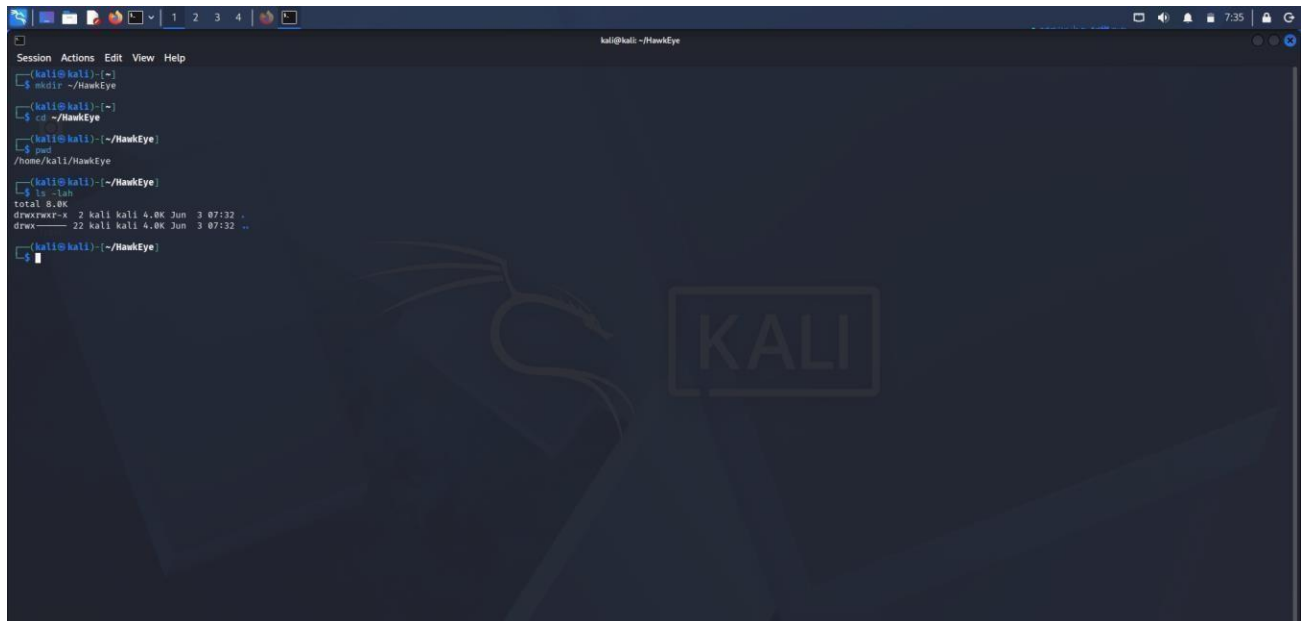
First, I created the folder using the command: **mkdir ~/HawkEye**

After creating the folder, I moved into it using: **cd ~/HawkEye**

To confirm that I was inside the correct directory, I checked the current location with: **pwd**

The output showed: **/home/kali/HawkEye**

Finally, I listed the contents of the folder using: **ls -lah**

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~/HawkEye'. The terminal shows the following commands and output:

```
(kali@kali)~$ mkdir ~/HawkEye
(kali@kali)~$ cd ~/HawkEye
(kali@kali)~/HawkEye$ pwd
/home/kali/HawkEye
(kali@kali)~/HawkEye$ ls -lah
total 8.0K
drwxrwxr-x  2 kali kali 4.0K Jun  3 07:32 .
drwx----- 22 kali kali 4.0K Jun  3 07:32 ..
```

The terminal background features a faint watermark of a hawk's head and the word 'KALI' in a box.

Step 5 - Moving the Lab file to the Project Folder

After creating the **HawkEye** project folder, I moved the downloaded lab file into it so that all project-related files would be stored in one place.

First, I navigated to the **Downloads** directory using the command: `cd ~/Downloads`

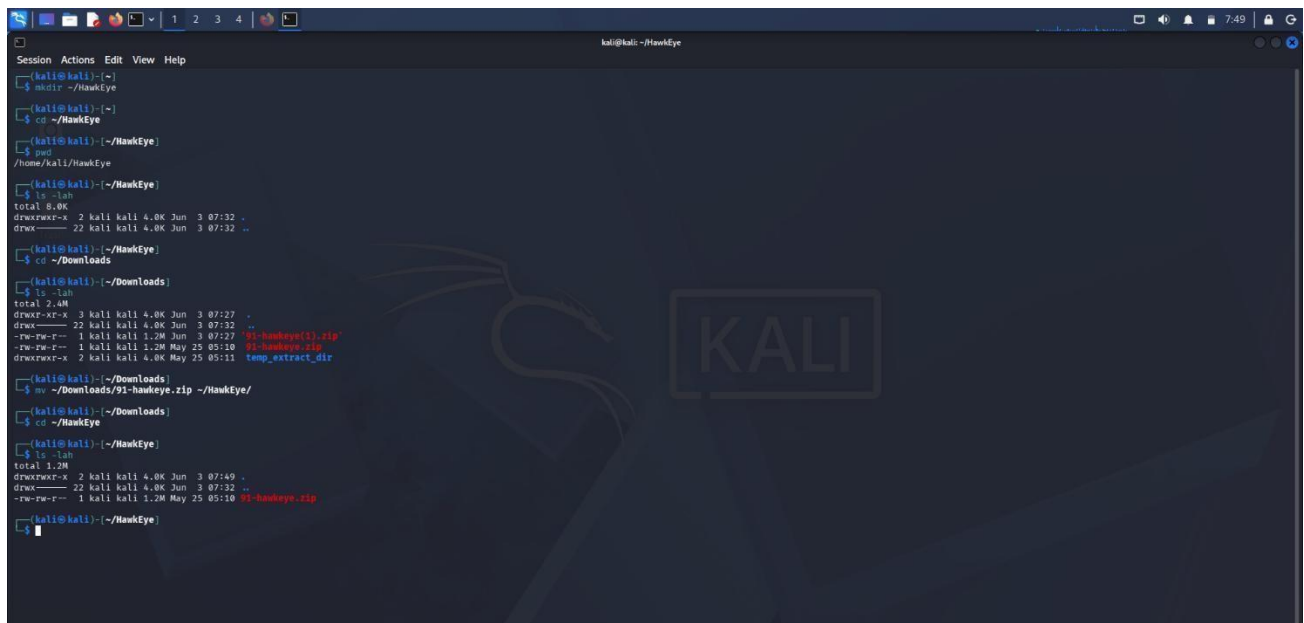
Then, I checked the available files by running: `ls -lah`

This command displayed the downloaded HawkEye ZIP file (**91-hawkeye.zip**) along with other files in the Downloads folder.

Next, I moved the HawkEye ZIP file from the Downloads folder to the HawkEye project folder using the command: `mv ~/Downloads/91-hawkeye.zip ~/HawkEye/`

After moving the file, I changed back to the HawkEye directory: `cd ~/HawkEye`

Finally, I verified that the file had been successfully moved by listing the contents of the folder: `ls -lah`



```
kali@kali: ~/HawkEye
Session Actions Edit View Help
❯ mkdir ~/HawkEye
❯ cd ~/HawkEye
❯ pwd
/home/kali/HawkEye
❯ ls -lah
total 8.0K
drwxrwxr-x 2 kali kali 4.0K Jun  3 07:32 .
drwx----- 22 kali kali 4.0K Jun  3 07:32 ..
❯ cd ~/Downloads
❯ ls -lah
total 2.4M
drwxr-xr-x 3 kali kali 4.0K Jun  3 07:27 .
drwx----- 22 kali kali 4.0K Jun  3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M Jun  3 07:27 91-hawkeye().zip
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
drwxrwxr-x 2 kali kali 4.0K May 25 05:11 temp_extract_dir
❯ mv ~/Downloads/91-hawkeye.zip ~/HawkEye/
❯ cd ~/HawkEye
❯ ls -lah
total 1.2M
drwxrwxr-x 2 kali kali 4.0K Jun  3 07:49 .
drwx----- 22 kali kali 4.0K Jun  3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
```

Step 6 - Extracting and Verifying the HawkEye Lab Files

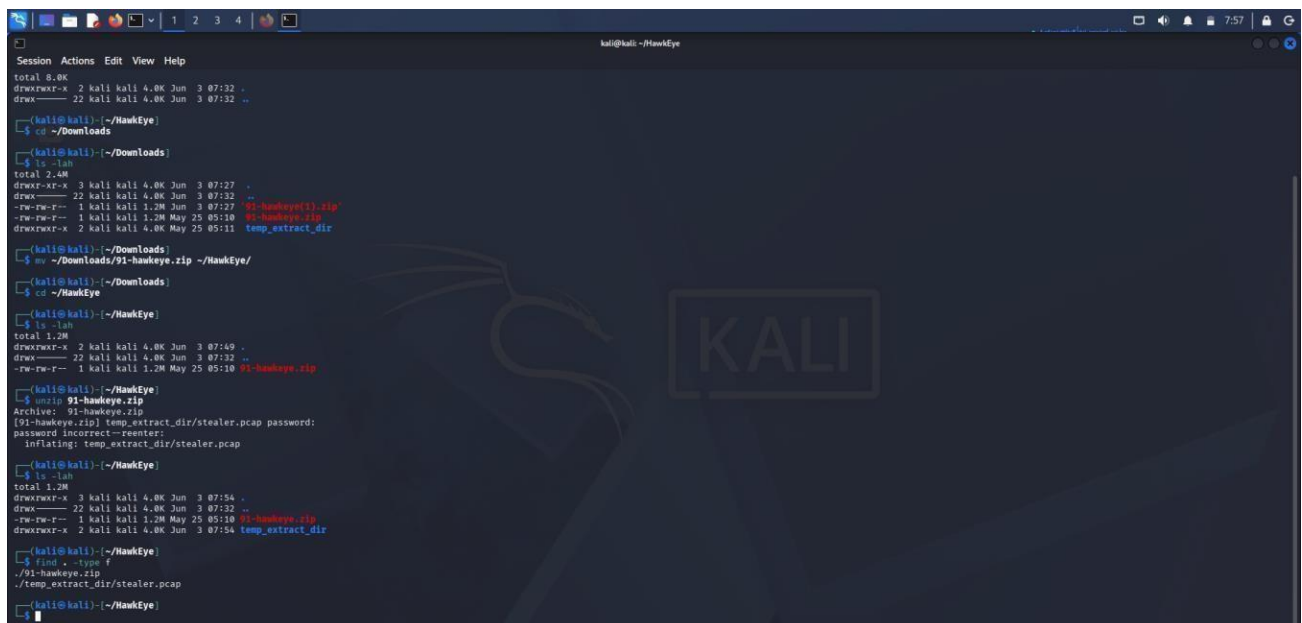
After moving the HawkEye ZIP file into the project folder, I extracted its contents so that I could access the investigation files required for analysis. I extracted the ZIP file using the following command: **unzip 91-hawkeye.zip**

After the extraction was completed, I checked the contents of the HawkEye folder again using: **ls -lah**

The output confirmed that a new directory named **temp_extract_dir** had been created.

Finally, I verified the extracted files by running: **find . -type f**

This command displayed all files inside the project directory and confirmed that the important network capture file **stealer.pcap** had been extracted successfully and was ready for investigation.



```
kali@kali: ~/HawkEye
Session Actions Edit View Help
total 8.0K
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:32 .
drwx----- 22 kali kali 4.0K Jun 3 07:32 ..

(kali@kali)~/HawkEye
$ cd ~/Downloads
(kali@kali)~/Downloads
$ ls -lah
total 2.4M
drwxr-xr-x 3 kali kali 4.0K Jun 3 07:27 .
drwx----- 22 kali kali 4.0K Jun 3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M Jun 3 07:27 91-hawkeye(1).zip
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
drwxrwxr-x 2 kali kali 4.0K May 25 05:11 temp_extract_dir

(kali@kali)~/Downloads
$ mv ~/Downloads/91-hawkeye.zip ~/HawkEye/
(kali@kali)~/Downloads
$ cd ~/HawkEye
(kali@kali)~/HawkEye
$ ls -lah
total 1.2M
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:49 .
drwx----- 22 kali kali 4.0K Jun 3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip

(kali@kali)~/HawkEye
$ unzip 91-hawkeye.zip
Archive: 91-hawkeye.zip
[91-hawkeye.zip] temp_extract_dir/stealer.pcap password:
password incorrect--reenter:
inflating: temp_extract_dir/stealer.pcap

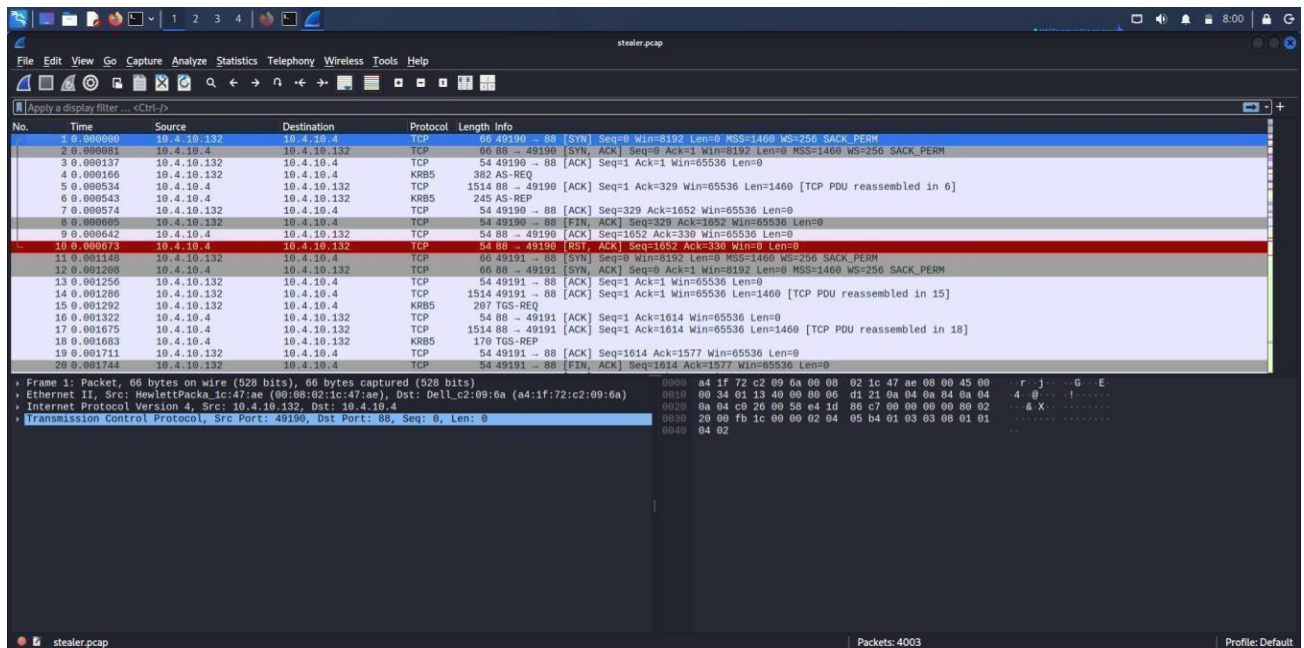
(kali@kali)~/HawkEye
$ ls -lah
total 1.2M
drwxrwxr-x 3 kali kali 4.0K Jun 3 07:54 .
drwx----- 22 kali kali 4.0K Jun 3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:54 temp_extract_dir

(kali@kali)~/HawkEye
$ find . -type f
./91-hawkeye.zip
./temp_extract_dir/stealer.pcap

(kali@kali)~/HawkEye
$
```

Step 7 - Opening the PCAP File in Wireshark

After extracting the HawkEye lab files, I opened the **stealer.pcap** file in **Wireshark** for network traffic analysis. Wireshark allowed me to inspect the captured packets and investigate the suspicious activities that occurred on the victim's system. Using Wireshark, I began exploring the network traffic to identify suspicious connections, malicious domains, email communications, stolen credentials and other indicators related to the HawkEye malware activity.

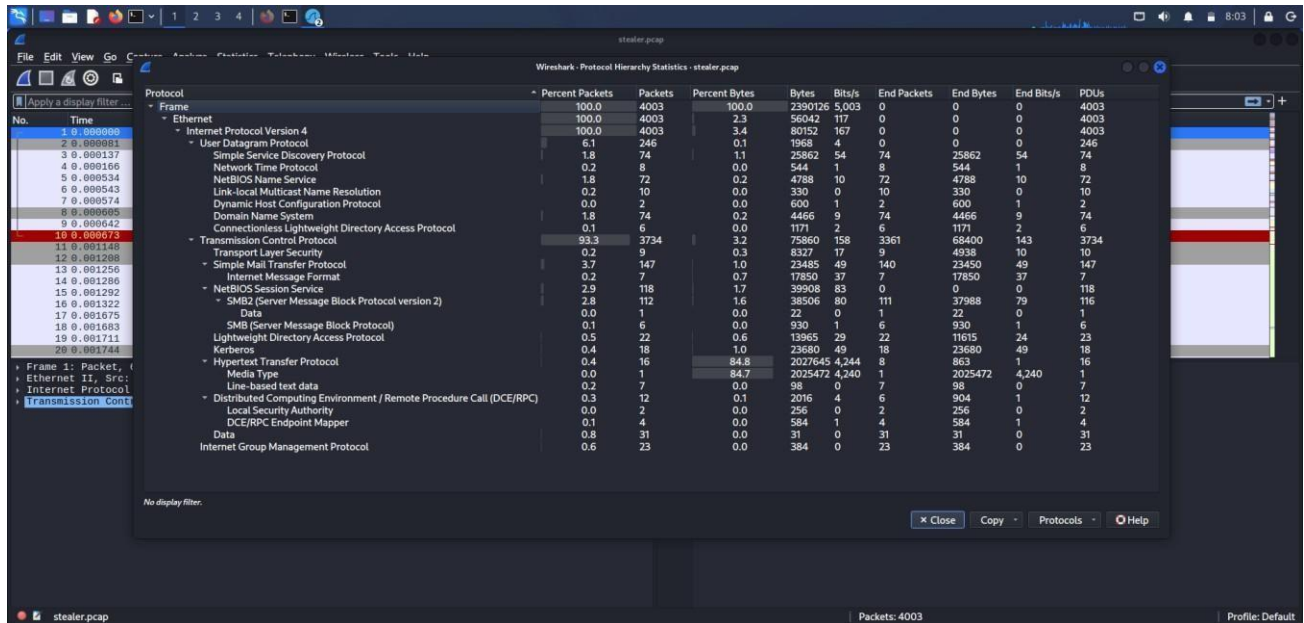


Step 8 - Checking Protocol Statistics in Wireshark

After opening the PCAP file, I used the **Protocol Hierarchy Statistics** feature in Wireshark to get an overview of the network traffic captured in the file.

To access this information, I selected: **Statistics** → **Protocol Hierarchy**

This window displayed the different network protocols present in the capture along with the number of packets and the amount of traffic generated by each protocol. I observed protocols such as **TCP**, **HTTP**, **SMTP**, **DNS**, **SMB2**, **LDAP**, **Kerberos** and **TLS**.



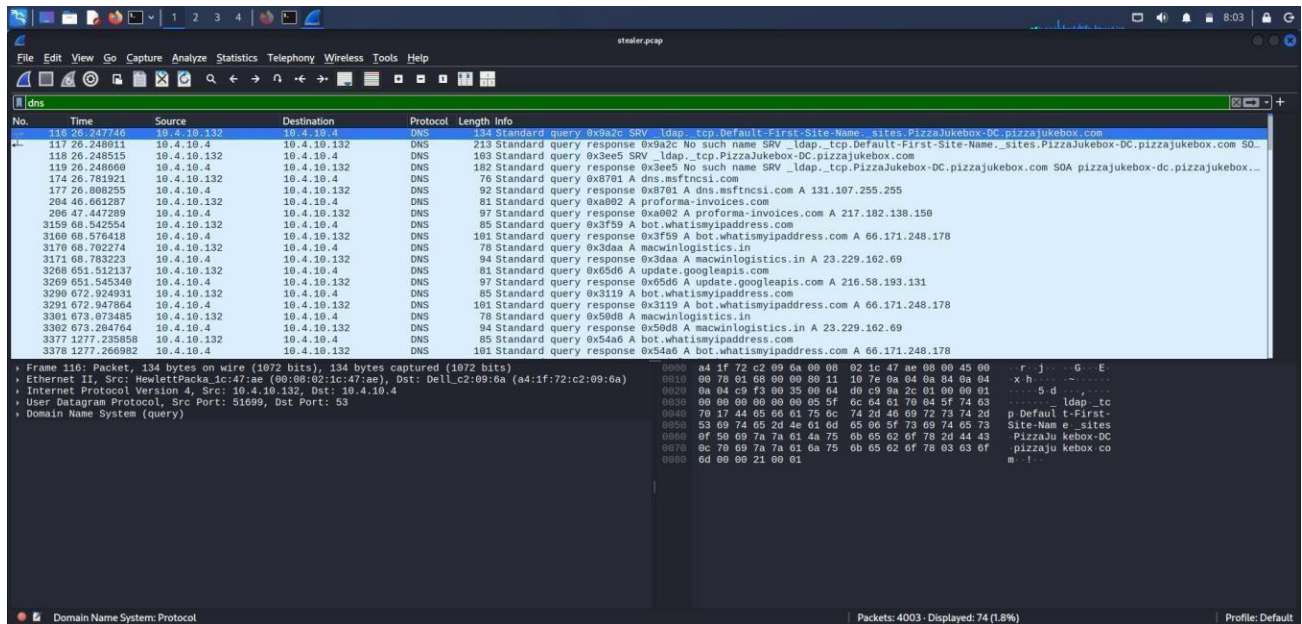
Step 9 - Analyzing DNS Traffic in Wireshark

To identify the websites and domains contacted by the infected system, I filtered the network traffic to display only DNS packets.

In the Wireshark filter bar, I entered: **dns**

After applying the filter, Wireshark displayed only DNS requests and responses.

I reviewed the domain names that were being queried by the system and noted several important domains, including external websites and servers contacted during the infection.



Step 10 - Analyzing HTTP Traffic in Wireshark

To find suspicious web activity, I filtered the network traffic to display only HTTP packets.

In the Wireshark filter bar, I entered: `http`

After applying the filter, I examined the HTTP requests and responses.

During the analysis, I found a GET request for a file named `tkraw_Protected99.exe` from the server 217.182.138.150.

The screenshot shows the Wireshark interface with the filter bar set to `http`. The packet list pane displays a list of HTTP packets. The selected packet (No. 210) is a GET request for `tkraw_Protected99.exe` from the server 217.182.138.150. The packet details pane shows the structure of the HTTP request, including the GET method, host, and user-agent.

No.	Time	Source	Destination	Protocol	Length	Info
210	47.837544	10.4.10.132	217.182.138.150	HTTP	392	GET /tkraw_Protected99.exe HTTP/1.1
3155	48.947474	217.182.138.150	10.4.10.132	HTTP	790	HTTP/1.1 200 OK (application/x-msdownload)
3164	60.640169	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3166	60.691423	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)
3295	673.609338	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3297	673.672155	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)
3382	1277.329651	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3384	1277.392746	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)
3467	1883.897476	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3469	1883.154667	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)
3582	2487.212975	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3584	2487.281176	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)
3837	3091.379849	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3839	3091.437608	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)
3915	3695.523251	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3917	3695.578849	66.171.248.178	10.4.10.132	HTTP	222	HTTP/1.1 200 OK (text/html)

Frame 210: Packet, 392 bytes on wire (3136 bits), 392 bytes captured (3136 bits) on interface 0
Ethernet II, Src: Hewlett-Packard, Dst: Netgear, Len: 150 (20:08:00:00:00:00) (20:e5:2a:b6:93:f1)
Internet Protocol Version 4, Src: 10.4.10.132, Dst: 217.182.138.150
Transmission Control Protocol, Src Port: 49204, Dst Port: 80, Seq: 1, Ack: 338
Hypertext Transfer Protocol

0000 20 e5 2a b6 93 f1 00 08 02 1c 47 ae 08 00 45 00G...E
0010 01 7a 01 a3 40 00 80 06 7f 06 0a 04 0a 84 d9 b6 ..z.@.....
0020 0a 90 c0 3a 00 50 a1 c3 4d 38 45 9a 09 5d 50 18 ...4P...MBE...JP
0030 7a f0 24 96 00 00 47 45 54 20 2f 70 72 6f 66 6f ...\$...GET/profo
0040 72 6d 61 2f 74 6b 72 61 77 5f 50 72 6f 74 65 63 rma/tkraw Protec
0050 74 65 64 39 39 2e 65 78 65 20 48 54 54 59 2f 31 ted99.exe HTTP/1
0060 2e 31 0d 0a 41 63 03 05 70 74 3a 20 2a 2f 2a 6d .1-Accept: */*
0070 0a 41 63 03 05 70 74 2d 45 6e 63 0f 64 69 6e 07 -Accept- Encoding
0080 3a 20 67 7a 69 70 2c 20 64 05 66 6c 61 74 65 8d : gzip, deflate
0090 0a 55 73 65 72 2d 41 07 65 6e 74 3a 20 4d 0f 7a User-Agent: Moz
00a0 69 6e 6e 61 2f 34 2e 30 20 28 63 0f 6d 70 61 74 illa/4.0 (compat
00b0 69 62 6c 65 3b 20 4d 53 49 45 20 37 2e 39 3b 20 ible; MS IE 7.0;
00c0 57 69 6e 64 6f 77 73 20 4e 54 20 36 2e 31 3b 20 Windows NT 6.1;
00d0 57 4f 57 36 34 3b 20 54 72 69 64 65 6e 74 2f 37 WOW64; Trident/7
00e0 2e 30 3b 20 53 4c 43 43 32 3b 20 2e 4e 45 54 20 .0; SLCC 2; .NET
00f0 43 4c 52 20 32 2e 30 2e 35 30 37 32 37 3b 20 2e CLR 2.0.50727;
0100 4e 45 54 20 43 4c 52 20 33 2e 35 2e 33 39 37 32 NET CLR 3.5.3072
0110 39 3b 20 2e 4e 45 54 20 43 4c 52 20 33 2e 30 2e 9; .NET CLR 3.0.
0120 33 3b 37 32 39 3b 20 4d 65 64 69 61 2b 43 65 6e 30729; Media Cen
0130 74 65 72 20 50 43 20 36 2e 30 3b 20 2e 4e 45 54 ter PC 6.0; .NET
0140 34 2e 30 43 3b 20 2e 4e 45 54 34 2e 39 45 29 8d 4.0C; .NET 4.0E)

Step 11 - Filtering HTTP Requests

To focus only on web requests made by the system, I applied the HTTP request filter in Wireshark.

In the filter bar, I entered: **http.request**

After applying the filter, Wireshark displayed only the HTTP requests sent from the system. I reviewed the requests and identified a download request for the file `tkraw_Protected99.exe`. I also examined the destination IP addresses and requested details to understand how the file was downloaded.

The image shows a Wireshark network traffic capture window. The filter bar at the top contains the text `http.request`. The packet list on the left shows a list of captured packets, with the first packet (No. 210) selected. The packet details pane on the right shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
210	47.597546	10.4.10.132	217.182.138.150	HTTP	392	GET /proforma/tkraw_Protected99.exe HTTP/1.1
3104	68.640169	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3244	109.882622	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3245	110.225538	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3247	112.899536	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3248	113.243393	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3249	115.915061	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3250	116.259007	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3295	673.085930	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3382	1277.329651	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3467	1883.097476	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3582	2487.212975	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1
3698	2671.938394	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3698	2674.969324	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3699	2676.429309	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3700	2676.469260	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3701	2679.489166	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3702	2679.529336	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3771	2863.169733	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1
3772	2863.208634	10.4.10.132	239.255.255.250	SSDP	175	M-SEARCH * HTTP/1.1

Frame 210: Packet, 392 bytes on wire (3136 bits), 392 bytes captured (3136 bits) on interface 0
Ethernet II, Src: Hewlett-Packard (08:00:27:1c:47:ae), Dst: Netgear (20:e5:2a:b6:93:f1)
Internet Protocol Version 4, Src: 10.4.10.132, Dst: 217.182.138.150
Transmission Control Protocol, Src Port: 49284, Dst Port: 80, Seq: 1, Ack: 1, Len: 338
Hypertext Transfer Protocol

Request: Boolean

Packets: 4003 | Displayed: 34 (0.8%)

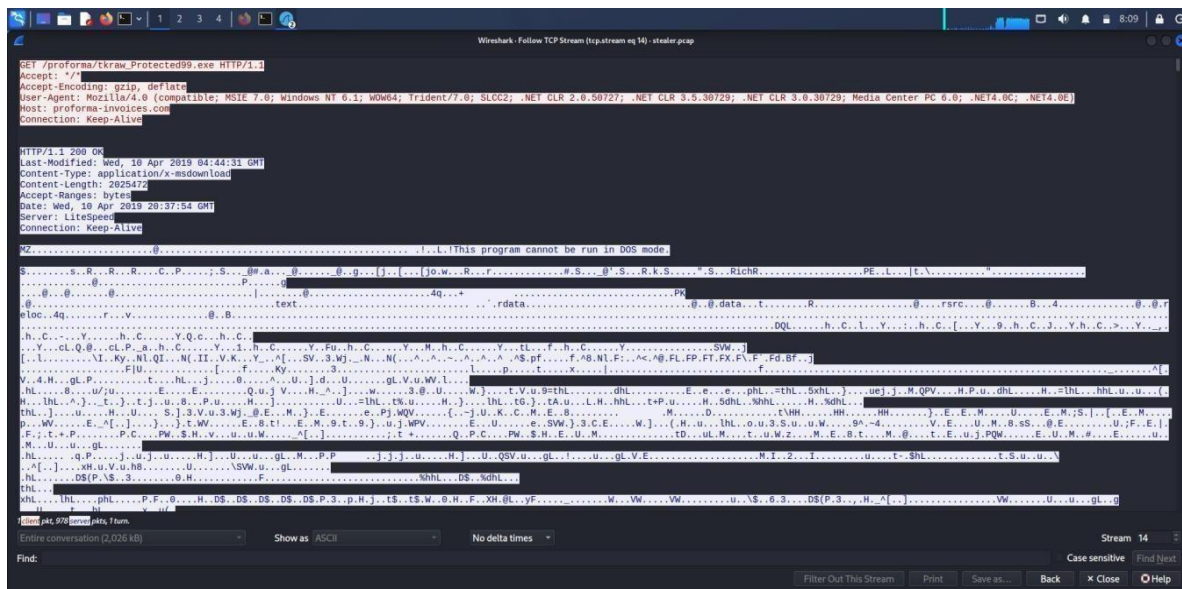
Profile: Default

Step 12 - Following the TCP Stream to Inspect the Downloaded File

To get more details about the downloaded file, I selected the HTTP packet related to **tkraw_Protected99.exe** and used the **Follow TCP Stream** option in Wireshark. This opened the complete communication between the infected system and the web server. In the stream, I observed the HTTP request, server response, file information, and the binary content of the downloaded executable file. The stream also showed details such as the file name, host name, content type, and file size. By examining this TCP stream, I confirmed that the system downloaded a suspicious executable file from the remote server. This helped me understand how the malware entered the system and provided important evidence for the investigation.

Key Findings:

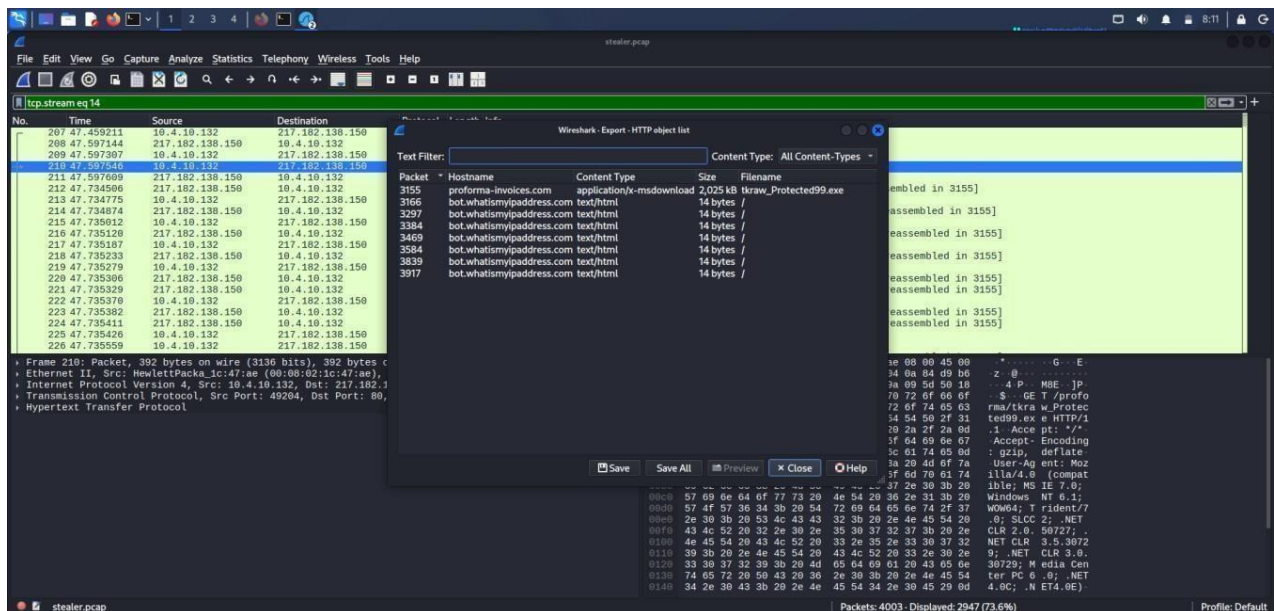
- Downloaded file: **tkraw_Protected99.exe**
- Host: **proforma-invoices.com**
- HTTP response: **200 OK**
- Content-Type: **application/x-msdownload**
- Executable file signature: **MZ**
- Download was completed successfully



Step 13 - Exporting and Reviewing HTTP Objects

After identifying the suspicious HTTP traffic, I opened File → Export Objects → HTTP in Wireshark to view all files transferred through HTTP communication.

The Export HTTP Objects window displayed a list of files that were exchanged between the infected system and external servers. In the list, I found a suspicious executable file named tkraw_Protected99.exe downloaded from proforma-invoices.com. The file size was approximately 2,025 KB, which confirmed that an executable file had been transferred to the victim's machine. I also observed several small HTTP responses from bot.whatismyipaddress.com, which were likely used by the malware to check the system's public IP address.

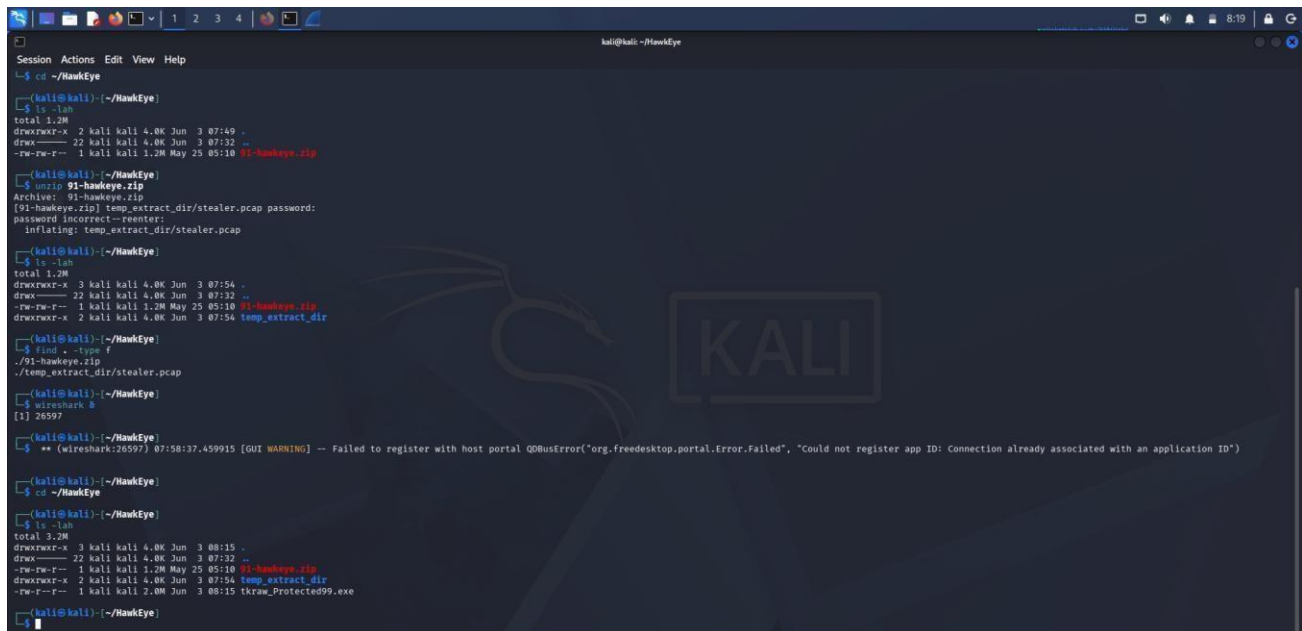


Step 14 - Saving the Suspicious Executable File

After identifying the malicious file in Wireshark, I saved the extracted file and verified that it was successfully stored in my project folder. To confirm this, I returned to the HawkEye directory and checked the available files using the `ls -lah` command.

The output showed a new file named **tkraw_Protected99.exe** along with the original ZIP file and extracted folder. This confirmed that the executable file had been successfully recovered from the network traffic and saved for further investigation.

During this step, I also used the `find . -type f` command earlier to locate all files inside the project directory and verify the location of the extracted evidence.



```
kali@kali: ~/HawkEye
$ cd ~/HawkEye
(kali@kali)~/HawkEye
$ ls -lah
total 1.2M
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:49 .
drwxr-xr-x 22 kali kali 4.0K Jun 3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
(kali@kali)~/HawkEye
$ unzip 91-hawkeye.zip
Archive: 91-hawkeye.zip
[91-hawkeye.zip] temp_extract_dir/stealer.pcap password:
password incorrect--reenter:
inflating: temp_extract_dir/stealer.pcap
(kali@kali)~/HawkEye
$ ls -lah
total 1.2M
drwxrwxr-x 3 kali kali 4.0K Jun 3 07:54 .
drwxr-xr-x 22 kali kali 4.0K Jun 3 07:32 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:54 temp_extract_dir
(kali@kali)~/HawkEye
$ find . -type f
./91-hawkeye.zip
./temp_extract_dir/stealer.pcap
(kali@kali)~/HawkEye
$ wireshark &
[1] 26597
(kali@kali)~/HawkEye
$ ** (Wireshark:26597) 07:58:37.459915 [GUI WARNING] -- Failed to register with host portal @BusError("org.freedesktop.portal.Error.Failed", "Could not register app ID: Connection already associated with an application ID")
(kali@kali)~/HawkEye
$ cd ~/HawkEye
(kali@kali)~/HawkEye
$ ls -lah
total 3.2M
drwxrwxr-x 3 kali kali 4.0K Jun 3 08:15 .
drwxr-xr-x 22 kali kali 4.0K Jun 3 07:32 ..
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:54 temp_extract_dir
-rw-r--r-- 1 kali kali 2.0M Jun 3 08:15 tkraw_Protected99.exe
(kali@kali)~/HawkEye
```

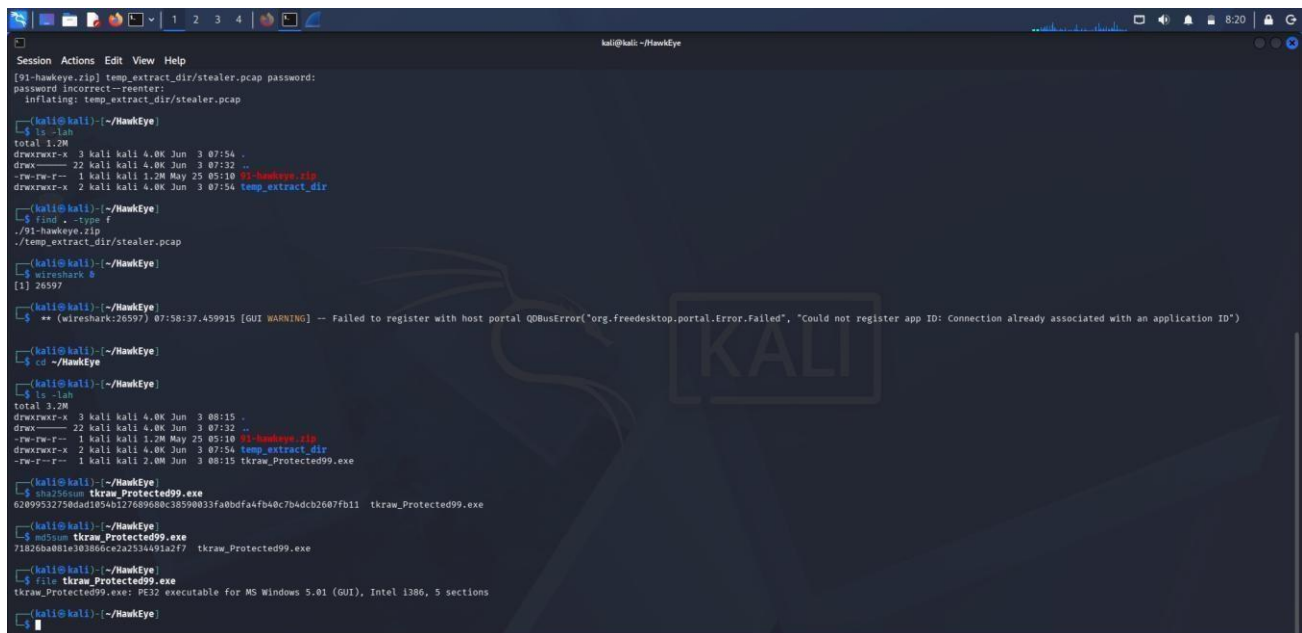
Step 15 - Calculating File Hashes and Identifying the File Type

After saving the suspicious executable file, I performed basic file analysis using Kali Linux terminal commands. The purpose of this step was to verify the file's integrity and determine its type before conducting any further investigation.

First, I used the `sha256sum tkraw_Protected99.exe` command to generate the SHA-256 hash value of the file. This created a unique digital fingerprint that can be used to identify the file and compare it with threat intelligence databases.

Next, I used the `md5sum tkraw_Protected99.exe` command to generate the MD5 hash value. Having both SHA-256 and MD5 hashes provided additional indicators that could be used during malware analysis and threat hunting activities.

Finally, I executed the `file tkraw_Protected99.exe` command to determine the file type. The output showed that the file was a **PE32 executable for Microsoft Windows**, indicating that it was a Windows application capable of running on Windows systems.



```
kali@kali: ~/HawkEye
Session Actions Edit View Help
[91-hawkeye.zip] temp_extract_dir/stealer.pcap password:
password incorrect - reenter:
inflating: temp_extract_dir/stealer.pcap

(kali@kali) ~/HawkEye
$ ls -lah
total 1.2M
drwxrwxr-x 3 kali kali 4.0K Jun 3 07:54 .
drwxr-xr-x 22 kali kali 4.0K Jun 3 07:52 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:54 temp_extract_dir

(kali@kali) ~/HawkEye
$ find -type f
./91-hawkeye.zip
./temp_extract_dir/stealer.pcap

(kali@kali) ~/HawkEye
$ wireshark &
[1] 26597

(kali@kali) ~/HawkEye
$ ** (wireshark:26597) 07:58:37.459915 [GUI WARNING] -- Failed to register with host portal QDBusError("org.freedesktop.portal.Error.Failed", "Could not register app ID: Connection already associated with an application ID")

(kali@kali) ~/HawkEye
$ cd ~/HawkEye

(kali@kali) ~/HawkEye
$ ls -lah
total 3.2M
drwxrwxr-x 3 kali kali 4.0K Jun 3 08:15 .
drwxr-xr-x 22 kali kali 4.0K Jun 3 07:52 ..
-rw-rw-r-- 1 kali kali 1.2M May 25 05:10 91-hawkeye.zip
drwxrwxr-x 2 kali kali 4.0K Jun 3 07:54 temp_extract_dir
-rw-r--r-- 1 kali kali 2.0M Jun 3 08:15 tkraw_Protected99.exe

(kali@kali) ~/HawkEye
$ sha256sum tkraw_Protected99.exe
62099532758dad1054b127689680c38598033fa8bdfa4fb48c7b4dc2607fb11 tkraw_Protected99.exe

(kali@kali) ~/HawkEye
$ md5sum tkraw_Protected99.exe
71826ba081e303866ce2a2534491a2f7 tkraw_Protected99.exe

(kali@kali) ~/HawkEye
$ file tkraw_Protected99.exe
tkraw_Protected99.exe: PE32 executable for MS Windows 5.01 (GUI), Intel i386, 5 sections

(kali@kali) ~/HawkEye
$
```


Step 16 - Viewing the Extracted File Content in the Terminal

After identifying and extracting the suspicious executable file, I examined its contents using the Kali Linux terminal. The file output displayed readable strings and section names that are commonly found inside Windows executable files.

While reviewing the output, I noticed the message "**This program cannot be run in DOS mode**", which is commonly found in Windows executable files. I also found sections such as `.text`, `.rdata`, `.data`, `.rsrc` and `.reloc`. These sections showed that the file was a Windows program and not a normal text file.

This step helped me verify that the recovered file was a genuine Windows executable and provided additional evidence that it could be the malware responsible for the suspicious activity observed in the network traffic.

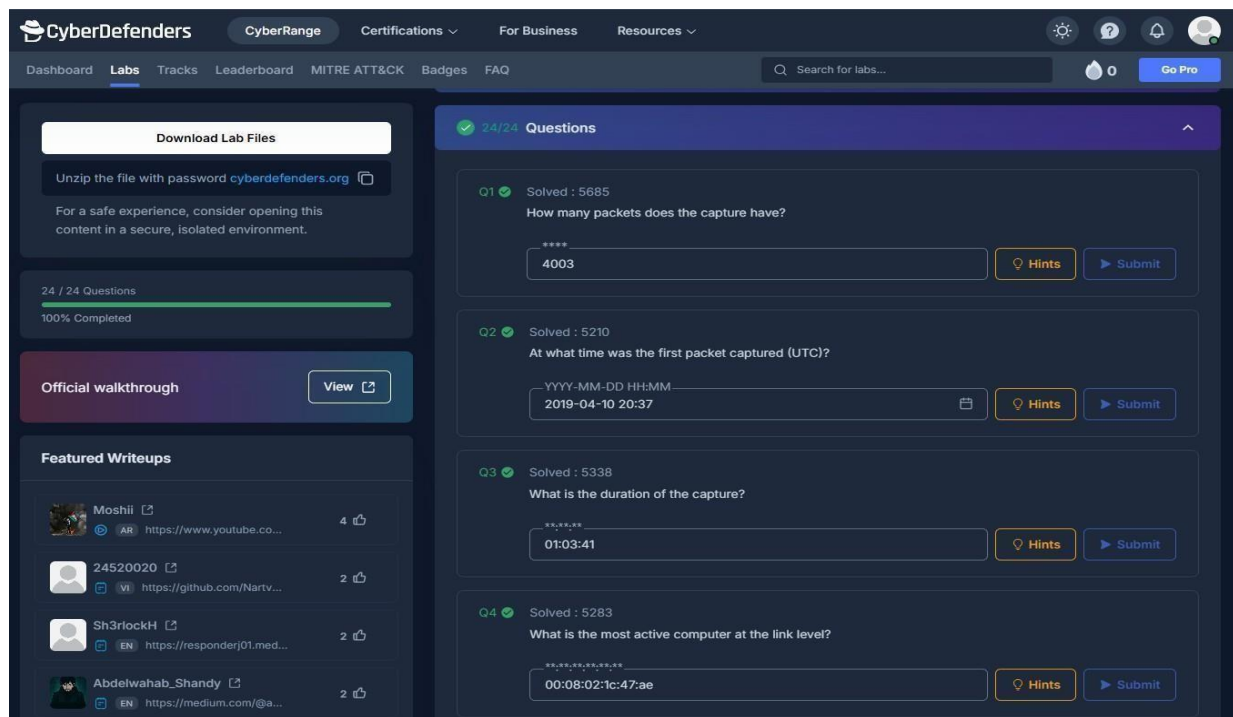
[illegible]

Step 17 - Investigating Network Traffic for Indicators of Compromise (IoCs)

After extracting and examining the downloaded file, I continued investigating the network traffic in Wireshark. I used different filters such as DNS, HTTP, SMTP, and TCP streams to view the communication between the infected system and external servers.

During the analysis, I identified suspicious domains, IP addresses, file downloads, and email communications related to the HawkEye malware. These findings helped me understand how the malware operated and how information was sent from the infected system to external destinations.

Step 18 - Solving the HawkEye Lab Questions



Using the evidence collected from the packet capture analysis, I answered all 24 questions provided in the HawkEye challenge. Each question requires examining different parts of the network traffic, including DNS requests, HTTP downloads, SMTP communications, malware hashes, credentials, and data exfiltration activity. I verified every answer using Wireshark and other analysis methods before submitting it on the CyberDefenders platform. After successfully answering all questions, the HawkEye challenge was completed.

Tools Used:

- Wireshark
- Kali Linux Terminal
- CyberDefenders Platform

Important Findings from the HawkEye Investigation:

Evidence Type	Finding
Malicious Domain	proforma-invoices.com
Public IP Check Domain	bot.whatismyipaddress.com
Downloaded File	tkraw_Protected99.exe
File Type	PE32 Windows Executable
HTTP Response	200 OK
Content Type	application/x-msdownload
Source IP	10.4.10.132
Download Server IP	217.182.138.150

SHA256 Hash:

62099532750dad1054b127689680c38590033fa0bdfa4fb40c7b4dcb2607fb11

MD5 Hash:

71826ba081e303866ce2a2534491a2f7

5. Skills and Competencies Gained

- Learned how to use **Wireshark** for network traffic analysis.
- Gained experience in analyzing **PCAP files**.
- Improved understanding of **DNS, HTTP, SMTP, and TCP** traffic.
- Learned how to identify suspicious network activities.
- Learned how to extract and examine suspicious files.
- Developed network forensic investigation skills.
- Learned how to identify **Indicators of Compromise (IoCs)**.
- Improved analytical and problem-solving skills during network investigations.
- Gained practical experience in malware investigation.

6. Feedback and Evidence

Feedback

- The HawkEye challenge provided practical experience in analyzing network traffic and investigating a malware incident.
- The challenge helped me understand how malicious files are downloaded and how stolen information can be sent to external servers through network communication.
- Using Wireshark improved my ability to analyze DNS, HTTP, and SMTP traffic and identify suspicious activities in a packet capture file.
- The hands-on investigation made it easier to understand network forensics concepts and how security analysts examine real incidents.
- Overall, the project was informative and helped improve my practical skills in network analysis and incident investigation.

Evidence Collected

- Packet capture (PCAP) file containing network traffic generated during the incident.
- DNS query records showing communication with different domains.
- HTTP requests used to download a suspicious executable file.
- SMTP traffic showing how information was sent from the infected system.
- Extracted executable file (tkraw_Protected99.exe) recovered from the network traffic.
- File hash values (MD5 and SHA256) generated for the extracted file.
- Network indicators such as IP addresses, domains, and URLs identified during the investigation.
- Answers to all 24 challenge questions based on the evidence found during the investigation.

7. Challenges and Solutions

During this project, I faced a few challenges while working on the HawkEye lab for the first time. Since I was new to this type of network forensic investigation, understanding the large amount of network traffic in the PCAP file was initially difficult. To overcome this, I learned how to use Wireshark filters such as DNS, HTTP and SMTP to focus only on the traffic that was important for the investigation.

Another challenge was identifying the suspicious activities hidden among thousands of packets. At first, it was difficult to determine which packets were related to the malware. I solved this by carefully examining the network traffic step by step and following the communication between the infected system and external servers.

I also faced challenges while extracting and analyzing the downloaded executable file. To understand the file better, I used terminal commands to generate file hashes and check the file type. This helped me confirm that the file was a Windows executable and an important part of the investigation.

Although these challenges required extra time and effort, they helped me improve my investigation skills, problem-solving ability and understanding of network traffic analysis. Working through these difficulties provided valuable practical experience and helped me complete the HawkEye lab successfully.

8. Outcomes and Impact

This project gave me valuable hands-on experience in network traffic analysis and cybersecurity investigation. Before starting the HawkEye lab, I had only a basic understanding of how network forensic investigations are performed. By completing this project, I learned how to examine network traffic and identify suspicious activities using practical tools and techniques.

Through the investigation, I was able to identify malicious domains, IP addresses, downloaded files, email communications and stolen credentials. I also learned how malware communicates with external systems and how sensitive information can be transferred from an infected computer to an attacker.

One of the main outcomes of this project was the improvement of my practical skills in using Wireshark and Kali Linux. I became more comfortable applying filters, analyzing packets, following network streams, and collecting evidence from captured traffic.

Overall, this project increased my understanding of network security, malware analysis, and digital forensics. It provided practical experience that strengthened my cybersecurity knowledge and helped me develop skills that will be useful in future learning and professional work.

9. Conclusion

This project provided me with an opportunity to gain practical experience in network traffic analysis and cybersecurity investigation. Throughout the HawkEye lab, I analyzed a packet capture file, investigated suspicious network activities, identified malicious communications, and examined how malware interacted with the infected system.

By using tools such as Wireshark and Kali Linux, I learned how to analyze different network protocols, track suspicious connections, recover important evidence and understand the techniques used by malware to steal and transfer sensitive information.

Overall, this project was a valuable learning experience that strengthened my understanding of cybersecurity concepts and provided hands-on experience with real-world investigation techniques. The knowledge and skills gained from this project will help me in my future studies and career in the field of cybersecurity.